Methodology

Analytical question

Can an NLP classifier identify products containing at least one substance from the supplied reference list?

Pipeline

1. Load and validate the labeled CSV.
2. Combine `ingredients_list` and product category into one text field.
3. Remove empty text rows.
4. Split data using a stratified 80/20 split with `random_state=42`.
5. Establish a majority-class baseline.
6. Transform text with TF-IDF using unigrams and bigrams.
7. Train balanced logistic regression.
8. Evaluate with accuracy, precision, recall, F1, ROC-AUC, average precision and confusion matrix.

Why recall?

For a screening tool, missing a product with a reference-based alert is more costly than producing a small number of additional alerts. Recall is therefore a key metric, but precision and calibration must also be monitored before any real deployment.